PYTHON MINI PROJECT

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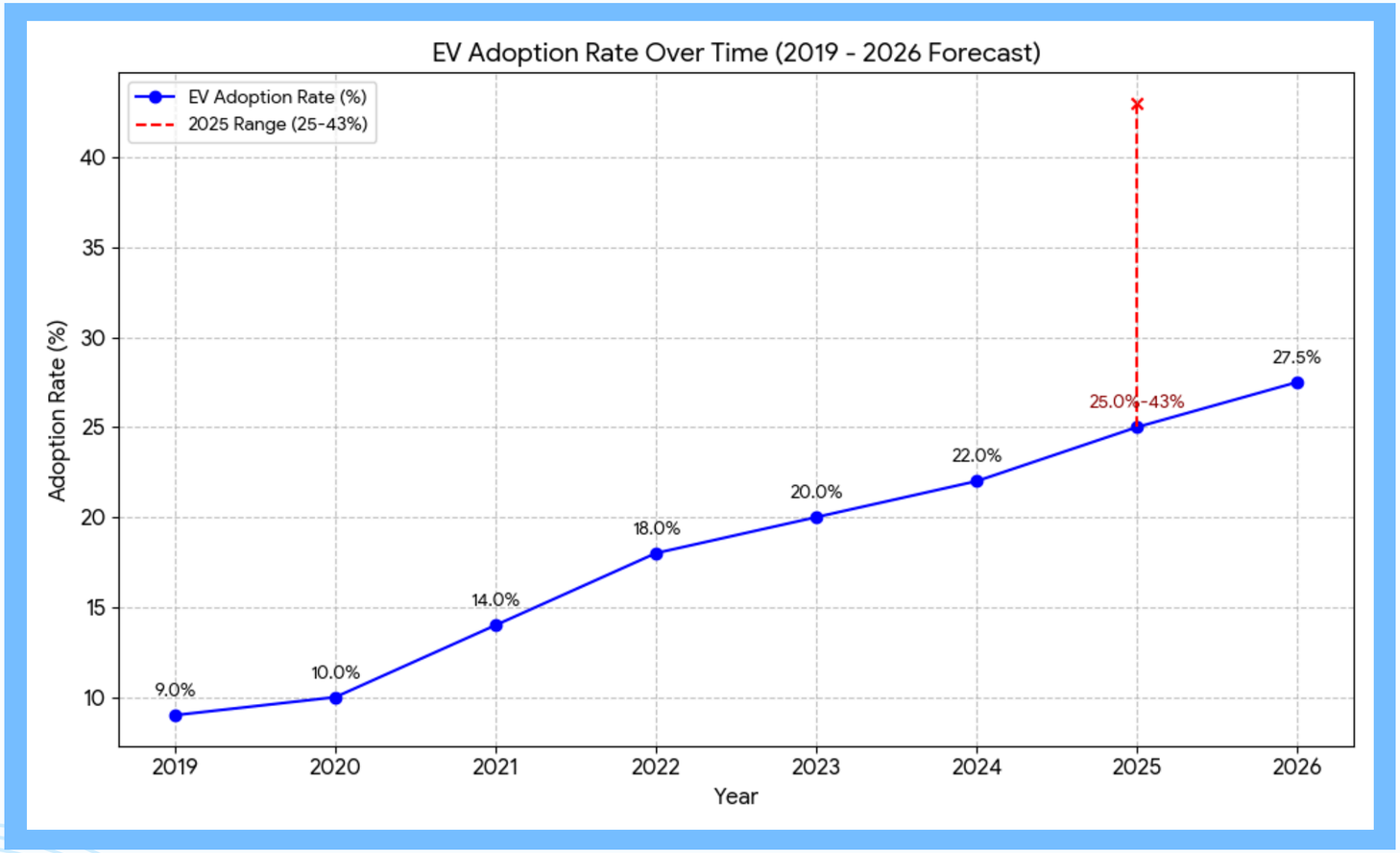
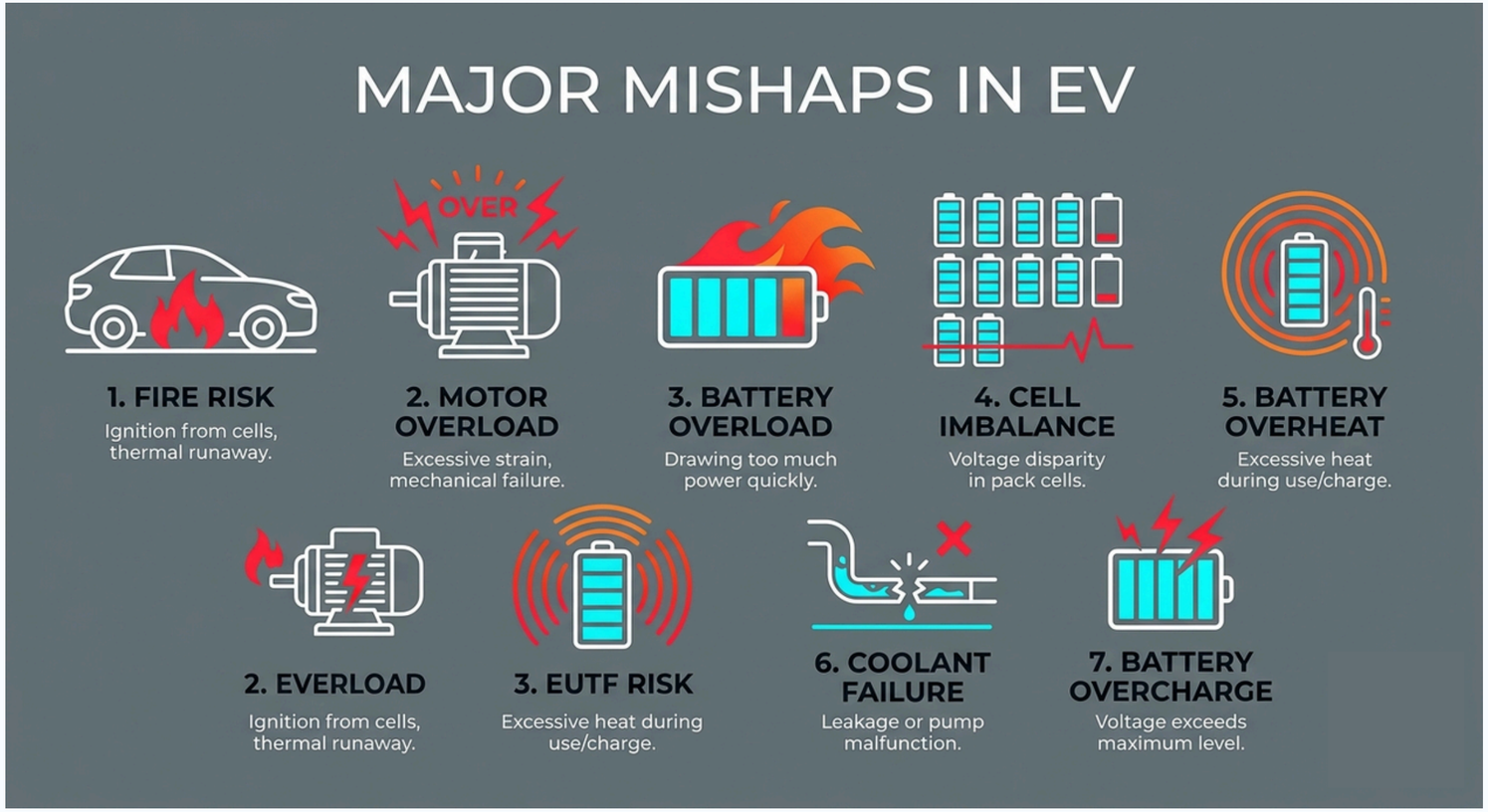
EARLY FAULT DETECTION IN EV

Reason for topic selection



PROBLEM STATEMENT



EARLY DETECTION DATASET

- We required dataset of observation of early behaviour of components before mishaps.

- So we have simulated dataset with 20 behaviour features and 2 categeorical column

- work flow: EDA -> preprocessing -> EDA verification ->train/test model -> predict

- python tool/libraries used : pandas ,numpy ,matplotlib ,seaborn ,scikit-learn

WEB WITH ML BACKEND

- Used Fastapi backend for this web page
- Early created .pkl trained model is now used with fast api backend
- This later pushed through git and made live using render serving domain
- It inputs 10 features and predicts fault label through trained model link : https://ev-web-kappa.vercel.app

CONSLUSION

- If these are implemented with accurate dataset we can implement it in actual vehicle sensor to monitor behaviour.
- This monitoring will display on dashboard any faults early so prevention measures can be taken.
- Therefore we can use it for human welfare.
- report link : my_report

THANK YOU